

Lab Worksheet 14
The Link Layer

The Link Layer, MAC addresses, Switches (aka Bridges), Hubs, Ethernet

This Week's Objectives

1. Be able to explain the differences between hubs and switches at the Link Layer.
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Expected Outcomes

After the laboratory session, students should be able to:

- explain the difference between a hub and a switch.
 - configure a software switch in Linux using brctl.
 - explain the different features of a captured pcap files, obtained from different bridge ports.
-

Workplan

These lab worksheets assume that you will make accurate notes of everything that you do for later consideration and revision. You should use your lab diary to note down items such as commands used, the outcome of each command, and answers to questions posed in these instructions.

All lab machines have CherryTree Notes software installed, which facilitates organised note-taking. Alternatively, you could buy a hardback A4 laboratory notebook.

NB this is a complex lab – please read each instruction carefully – if you miss out any steps, then don't expect to achieve the correct results! You will need to have completed and thoroughly understood all the steps before the next lab session.

Remember that there is a lab work wiki available to you on BB. Please use this to share and communicate with your colleagues.

Make sure you have completed all the steps in this worksheet before the next lab session.

Lab Worksheet 14
The Link Layer**Starting assumptions**

1. You have completed all the work in previous lab worksheets
 - You are familiar with basic Linux commands used to inspect and manipulate directories and files.
 - You have used *Netkit* (specifically *vstart*) to launch two virtual machines.
 - You have used *ip* commands within the virtual machines to configure network interfaces.
 - You have saved captured network traffic from a virtual machine using *tcpdump*.
 - You have viewed a packet capture file in the real Linux host using *wireshark*.

Activities

Reminder

- You are prepared to make accurate notes of what you do
- You will complete unfinished activities before the next timetabled lab session.
- You will resolve what you do not understand by conducting your own careful (and ethically sound) experimentation and/or further reading.

Getting Started

2. Create a directory for this lab's work as per previous labs. Use a relevant directory name.
3. Download the two netkit lab zipped archives (*netkit_lab_14_a.tar.xz* and *netkit_lab_14_b.tar.xz*) from the Blackboard shell and save them in your weekly directory. Each archive contains the configuration information for multiple netkit virtual machines.
4. Unzip the file *netkit_lab_14_a.tar.xz* with the command:

```
.../lab_14$ tar -xvf netkit_lab_14_a.tar.xz
```

5. Move into the new directory and list the directory with the *ls* command. You will see files and directories to create 4 xterms.

```
.../lab_14$ cd netkit_lab_14_a
.../netkit_lab_14_a$ ls -l
```

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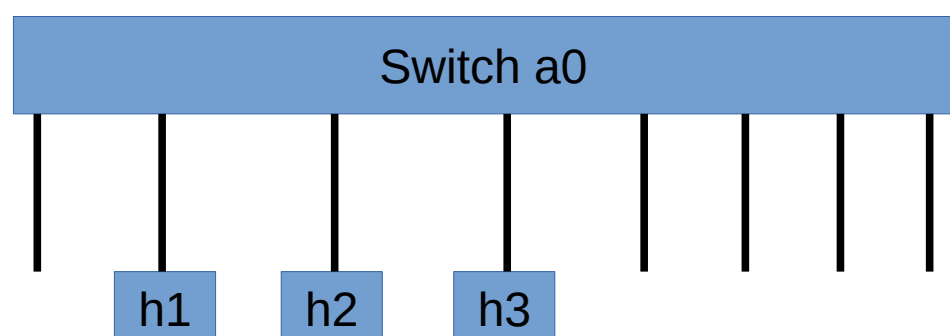
6. You will see
- one lab.conf file
 - one lab.dep file
 - four directories (all empty)
 - four *.startup files
7. Using the tree command:

```
.../netkit_lab_14_a$ tree
```

```
.
├── a0
├── a0.startup
├── h1
├── h1.startup
├── h2
├── h2.startup
├── h3
├── h3.startup
├── lab.conf
└── lab.dep

5 directories, 6 files
```

8. The network simulates the configuration of a network switch/bridge (a0) and three hosts (h1, h2, h3). The switch (a0) will not have a network (IP) address. Each of the three hosts will have a network address so we can use them to ping each other. We have given the switch (a0) eight ports, connected three of them to h1, h2, and h3, and left the other five unused.
9. Connectivity is as in the diagram below (draw this in your own notes to annotate as the lab progresses):



Lab Worksheet 14
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Use the *cat* command to review the contents of a file. NB only ever open a file in a text editor if you mean to make some changes to it.

Eg:

```
cat lab.conf
```

What machines will be added to the switch, and how is this done?

***.startup files**

10. Inspect the contents of each of the startup files.
11. These files contain the final instructions that are executed, before you are presented with the login prompt of the virtual machine. The startup files for h1, h2, and h3 are all very similar, and they are only two lines long. Look at the content of each of these files in turn and use the content to annotate your network diagram in your notes. Ensure you get the network card designation (eth0, eth1 etc), the MAC address and the IP address recorded accurately.
12. The file *a0.startup* is more complex. This represents a *switch* (or bridge – essentially the same thing). Note the MAC addresses allocated to each Ethernet card. What do you notice?
13. What is the name that we give to the bridge using the *brctl* instruction?
NB The switch is configured and managed with the Linux *brctl* command (*man brctl*).
14. What instruction is used to bond an interface card (eth4 for example) to the bridge?
15. Do any of the network devices on a0 have an IP address?
16. With your current directory as the *netkit_lab_14_a* directory (which contains the *lab.conf* and *lab.dep* files), launch the four virtual machines using the instruction:

```
.../netkit_lab_14_a$ lstart
```

17. You will see four virtual machines start up in quick succession (a0 will be slower than the other three). Once they have started, move them to positions on the screen that match the diagram.

Lab Worksheet 14
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18. On the *a0* machine, look at the configuration of all the network devices. Explain the relationship between the MAC address of the switch (*sw0*) and the *ethn* devices.

```
root@a0:~# ip link
```

19. Look at what the switch already knows about MAC addresses so far (record this in your notes):

```
root@a0:~# brctl showmacs sw0
```

Output:

NB the switch will collect MAC addresses from all messages that pass through it.

```
root@a0:~# brctl showmacs sw0
port no mac addr          is local?    ageing timer
 2      02:60:ac:9a:bc:01    no           31.34
 3      02:60:ac:9a:bc:02    no           36.44
 4      02:60:ac:9a:bc:03    no           21.12
 1      08:00:4e:a0:0a:00    yes          0.00
 1      08:00:4e:a0:0a:00    yes          0.00
root@a0:~#
```

20. On *h2*, use *tcpdump* to capture all traffic on its *eth0* device. NB the directory */hosthome* corresponds to the root directory of your logged in user account.

```
root@h2:~# tcpdump -i eth0 -s0 -w
/hosthome/ctec1704/captures/h2-d1.pcap
```

21. Similarly, on *h3*, use *tcpdump* to capture all traffic on its *eth0* device.

```
root@h3:~# tcpdump -i eth0 -s0 -w
/hosthome/ctec1704/captures/h3-d1.pcap
```

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22. Now on *h1*, we are going to ping *h3* with precisely one ping (technically this is at the network layer but is the least intrusive way of getting generating a tiny amount of link layer traffic).

```
root@h1:~# ping -c1 192.168.97.43
```

23. Stop the *tcpdump* in *h2* and *h3* using *ctrl-c*. Note in both cases how many packets (technically these are actually frames, not packets) have been received by the filter.
24. Look at what the switch now knows about *mac* addresses. On *a0* execute the below command. What do you notice?

```
root@a0:~# brctl showmacs sw0
```

25. On the actual host system, use *wireshark* to view *h2-d1.pcap* and *h3-d1.pcap*. In each case, explain what you can see there in the context of what the switch knows of the MAC addresses to which it is connected.
26. On *h2*, ping *h3* precisely twice:

```
root@h2:~# ping -c2 192.168.97.43
```

27. Look again at what the bridge knows, using:

```
root@a0:~# brctl showmacs sw0
```

28. Explain what you see. In particular:
- What does the switch use to learn which MAC addresses can be found through which port?
 - What does the switch do when it does not know which port to use for a MAC address?
29. Now to expand the network:

Lab Worksheet 14
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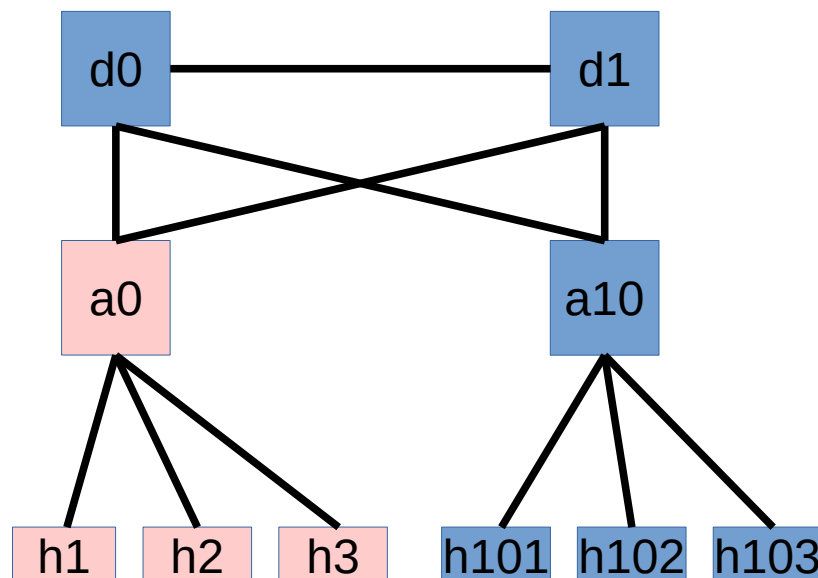
30. Starting assumption: the four machines of *netkit_lab_14_a* (ie a0, h1, h2, h3) are running.
31. Move into the *lab_14* directory, and extract the second example file.

```
.../lab_14/netkit_lab_14_a$ cd ~/ctec1704/lab_14
.../lab_14$ tar -xvf netkit_lab_14_b.tar.xz
```

32. Open a second terminal (hint: ctrl shift t will open a new terminal tab), move into the new directory *netkit_lab_14_b* and list its contents.

```
.../lab_14/$ cd netkit_lab_14_b
.../lab_14/netkit_lab_14_b$ ls -l
```

33. This directory contains the configuration information for six further virtual machines. This represents the configuration of three more switches (*a10*, *d0* and *d1*) and three new hosts (*h101*, *h102*, *h103*).
- The switches will not have network (IP) addresses.
 - Each of the three hosts will have a network address so we can use them to ping each other.
 - The switch (*a10*) has been configured with eight ports, connected three of them to *h101*, *h102*, and *h103*, connected one to switch *d0* and one to switch *d1*.
 - Review the *lab.conf* file to understand the connections.
 - Any remaining ports are unused.
 - Connectivity is as in the diagram below (draw this in your own notes to annotate as the lab progresses):



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34. Note the presence of redundant links between *a0*, *a10*, *d0* and *d1*. These extra links create alternative paths between hosts.

For example a message from *h1* to *h102* could be routed by any of these:

h1 => *a0* => *d1* => *a10* => *h102*

h1 => *a0* => *d0* => *a10* => *h102*

h1 => *a0* => *d1* => *d0* => *a10* => *h102*

etc

In a worst case scenario, messages could simple cycle around the switches and possibly never arrive at their destination – know as the looping problem

Fortunately another protocol - the [Spanning Tree Protocol](#) (STP) - was developed to prevent looping problems.

35. Note the six directories (all empty), one for each virtual machine.
36. Launch the new lab using:

```
.../netkit_lab_14_b$ lstart
```

37. Go to each of the four switches and, after each visit, display the MAC addresses each has learned. Write these on your diagram. Contrast this to the situation where we only had the one switch *a0*.

```
root@d0:~# brctl showmacs sw0
```

38. *ping* *h3* from *h101*

```
root@h101:~# ping -c1 192.168.97.43
```

39. then use *brctl* again to show what each switch has now learned about MAC addresses.

```
root@d0:~# brctl showmacs sw0
```

40. What can you infer about which switches carried the ping traffic.

Lab Worksheet 14
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41. Turn on tcpdump, listening on *d1*

```
root@d1:~# tcpdump -s0 -i sw0 -w /hostlab/d1-dump1.pcap
```

42. Bring down selected links between the switches, for example on *d0* try executing:

```
root@d0:~# ip link set eth3 down
```

Spanning Tree Protocol

Review: From the diagram above, you will remember that multiple routes exist between hosts:

eg

h1 => a0 => d0 => d1 => a10 => h103

or

h1 => a0 => d1 => d0 => a10 => h103

This can lead to broadcast storms, where messages can loop continuously. To prevent this, the Spanning Tree Protocol was developed. Read about this [here](#) and [here](#).

43. Bring all the links back up, then disable spanning tree on all the switches:

```
brctl stp sw0 off
```

44. Make sure you have tcpdump running on one of the switches, then ping a non-existent host. See if you can achieve a broadcast storm.
45. Experiment!!!